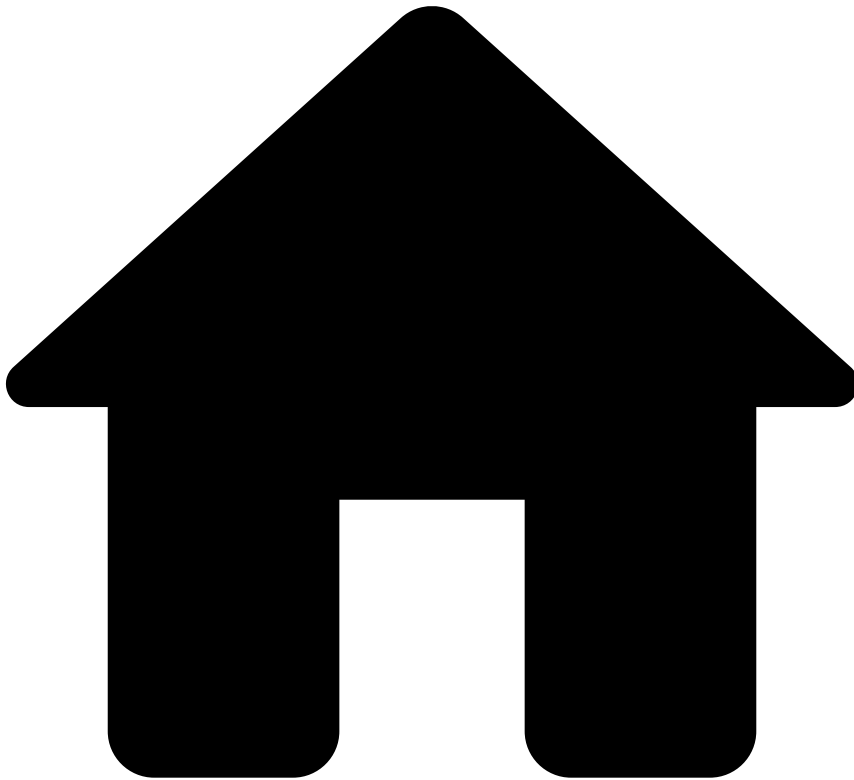


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Set Up Microsoft LightGBM Logging

Was this page helpful? 

To log information about what happens when training a model with the LightGBM provider, you can configure the logging levels from the most basic "INFO", up to the most extreme level "TRACE".

Logging levels

The LightGBM provider has several modules, each with different responsibilities. You can control the logging level per module.

Module name	Description	Logging level description
LightGBMBinaryClassificationModel	Implements the OpenML class responsible for loading a model and scoring instances. It is also called when saving the model to disk.	ERROR: Report that saving model to disk failed.
LightGBMBinaryClassificationModelTrainer	Responsible for training a LightGBM model. At the end of the train it saves an intermediate representation to disk that is then passed to LightGBMBinaryClassificationModel for the final model save to disk.	• INFO: Information messages reporting the stage at which the training is. • ERROR: For errors during train. • DEBUG: Report #train-instances, training parameters and features, more details on the sub-stages of training, progress in number of train iterations.
LightGBMModelCreator	Starts all load/fit operations and validations prior to starting the aforementioned model load/fit process.	• INFO: Message with filepath from which the model is being loaded. • ERROR: Errors when training the model or the validations for a model fit/load fail and where it is it trying to read from.
LightGBMSWIG	Low-level implementation for model scoring.	• INFO: Saving model to disk calls. • ERROR: Errors saving model to disk. • DEBUG: Saving model to disk calls with file paths. • TRACE: Show prediction score per instance. Note that this level is very verbose!
LightGBMUtils	Loads the LightGBM libraries.	• INFO: When the libraries are loaded. • DEBUG: Names of the libraries being loaded.
SWIGResources	Lowest-level implementation level for resources handling during model scoring.	• DEBUG: In-memory instantiation of model. • TRACE: Resource release calls.
SWIGTrainResources	Lowest-level implementation level for resources handling during model fitting.	DEBUG: Details for train memory allocation.

How to set up logging🔗

To configure such log levels, edit the file `conf/logger.xml`. Also, for Spark Standalone `spark-lib/logback-executors.xml` should also have such log levels.

Example configuration🔗

An example of configuring for maximum log level follows. Feel free to copy-paste, but beware of having too much logging enabled (see table above):

`conf/logger.xml` & `spark-lib/logback-executors.xml`

```
<logger name="com.feedzai.openml.provider.lightgbm.LightGBMBinaryClassificationModel" level="trace"/>
<logger name="com.feedzai.openml.provider.lightgbm.LightGBMBinaryClassificationModelTrainer" level="trace"/>
<logger name="com.feedzai.openml.provider.lightgbm.LightGBMModelCreator" level="trace"/>
<logger name="com.feedzai.openml.provider.lightgbm.LightGBMSWIG" level="trace"/>
<logger name="com.feedzai.openml.provider.lightgbm.LightGBMUtils" level="trace"/>
<logger name="com.feedzai.openml.provider.lightgbm.SWIGResources" level="trace"/>
<logger name="com.feedzai.openml.provider.lightgbm.SWIGTrainResources" level="trace"/>
```

